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Factors influencing audit quality through the acceptance and use of AI by auditors in Thailand, affiliated with SMEs audit firms

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***Abstract*—This study investigates how technology and organization factors related to the technology organization and external (TOE) model (technological, organizational, and environmental) affect how auditors perceive and use artificial intelligence (AI) tools and the resulting quality of audits performed. The study also uses the technology acceptance model (TAM) to assess audit quality resulting from the use of AI tools, using survey data collected from 120 auditors at small and medium sized companies in Thailand. The data was analyzed using basic stats and structural equation modelling (partial least squares - structural equation modelling (PLS-SEM) with ADANCO 2.4.1 and 5000 bootstrapping iterations). Results indicate that the measurement tool reached acceptable levels of reliability and convergent validity. The structural model's results linked the independent variables to significant variance in perceived AI benefits (73.73%) and AI acceptance and use (71.24%). The variance in the dependent variable of perceived ease, usefulness, and audit quality used AI tools were very low. Hypothesis tests showed that there was a statistically significant and positive relationship between environmental factors and perceived benefits of AI based on legal regulations ($\beta = 0.562$, $p < .05$) while all other independent variable factors were estimated to have no statistically predictable relationship with perceived benefits of using AI for auditing. Thus, the findings provide useful information for auditors regarding the effects on audit success of adopting new technology. These findings can also aid organizations, regulatory agencies, and the Federation of Accounting Professions in developing a plan to support the use of new technologies.**

Keywords*—technology-organization-environment framework, technology acceptance model, artificial intelligence, auditing, audit quality

I. INTRODUCTION

Currently, the industrial sector in Thailand is preparing in all aspects to accommodate technological changes that affect work systems, data storage, and service delivery, to promote and facilitate organizations and increase work efficiency [1]. According to regulations, all industrial entities in Thailand are required to prepare financial statements and income statements for auditing and presentation to an auditor for approval at a meeting during every accounting period [2]. The purpose of auditing is to express an opinion that the financial statements are fairly accurate in all material respects in accordance with generally accepted accounting principles. Auditing requires obtaining sufficient and appropriate evidence to achieve reasonable assurance in expressing an opinion [3]. With the transition of legal entities to digital operations, there is an increase in the volume of data being generated and stored, creating more complex data structures. Furthermore, traditional methods used for auditing are no longer adequate in providing sufficient and pertinent evidence for the purposes of auditing; therefore, there is a need to augment auditing processes through the application of Artificial Intelligence (AI).

The use of AI has greatly enhanced how audits are performed by enabling auditors to analyze large amounts of information at a rapid speed, which improves the accuracy in identifying risks and anomalies and providing better evaluation of audit evidence that can lead to higher quality audits; however, auditors can only maximize the advantages of using such technology if they understand what influences their level of acceptance and utilization of it. Therefore, this study is designed to examine the factors that influence audit quality in the way that auditors in Thailand utilize AI as a source of support, especially for those who belong to medium and small audit firms. In addition to identifying these factors using a technology, organization and environment framework, the study will also consider how technology acceptance theory affects the factors identified in the study. The results of this research will allow audit firms, auditors and the Federation of Accounting Professions to develop strategies to improve the way that auditors use AI.

II. RELATE WORK

A. AI in Auditing

The audit standards set forth by The Federation of Accounting Professions (TFAC) consist of four components: Audit Plan [3], Risk Assessment [4], Internal Control Testing [3,5] and Content Audit [5]. These principles must be adhered to during the audit process. The use of manual auditing across an expansive data set results in errors and delays. The Auditing Profession must move towards "real-time auditing" [6] from "retrospective auditing" as the service changes due to the introduction of IOT, Big Data, and other key technologies. By utilizing things like Blockchain [7], RPA, Machine Learning [8], Big Data, Cloud Computing, and Deep Learning [9], the efficiency of audit processes will improve. The application of these technologies will not only eliminate/reduce redundancy and time in regards to the complexity of the business [10-11], but these technologies will also enhance the accuracy/reliability of Financial Statements [10,12]. These findings should lead to a significant cost savings from the reduction of audit personnel [10] required to perform the audit work.

The application of artificial intelligence (AI) is improving both the predictive and analytical skills of auditors as they work on areas such as financial forecasting and operational assessments [13] , while natural language processing (NLP) tools assist with document extraction. This allows auditors more time to perform strategic tasks at a higher level [14]. As well, big data analytics support auditors in detecting fraud and help limit reliance on sampling when assessing internally controlled processes [15]; moreover, as big data creates a framework for auditors to assess their client's overall strategy and digital transformation [16], it also has a positive influence on IT governance by providing auditors with the ability to build continuously automated auditing systems that meet business and legal standards [17].

Advancements in technology will continue to force auditors to constantly investigate and improve their data analytic skills [11] by applying emerging technologies (including big data, cloud, Industry 4.0 [9], Machine Learning, blockchain and robotics process automation) throughout the audit process (planning, execution and reporting). The integration of these technologies reduces the potential for error, saves time, improves effectiveness and enhances the value provided by an audit firm.

B. Audit Quality

According to International Auditing and Assurance Standards Board (IAASB) [18], auditors must indicate whether the financial statements are materially misstated based on sound and appropriate evidence. Such audits must follow specified Standards to ensure that users of the financial statements have the highest degree of assurance.

The quality of an audit consists of three components: 1) Inputs – consist of the qualities of the auditors themselves (ethics, integrity, independence, and competence) as well as the way the audit team manages and understands the clients' business under the audit firm's quality-control policies and systems (International Standards for Quality Control 1 and 2), 2) Procedures – consist of those procedures required by the IAASB [19-20] in relation to compliance with standards and laws, including planning, risk assessment, evidence-gathering, application of technology, and continuous improvement through communication with clients and 3) Outcomes – of the audit are reported in the form of an unqualified audit report that demonstrates an audit's ability to meet the needs of its users by instilling confidence, furthering internal controls, enhancing financial reporting processes, providing useful information to stakeholders to support their decision-making processes, and ultimately creating business value.

Quality inspection is based on inputs, standard procedures, and the use of technology for the creation of reports that are complete, trustworthy, and useful to all interested parties for making decisions.

C. Technology Acceptance Model (TAM)

In 1989, Fred Davis proposed the Technology Acceptance Model (TAM) that attempts to describe and predict whether or not people will utilize new technology. There are two basic drivers to deciding to accept/use technology - perceived benefits of enhancing efficiency/effectiveness through use of the technology, and perceived ease of use of the technology. Both of these drivers are influenced by other outside factors that ultimately impact user's perceptions and ultimately their intentions towards using the technology.

TAM evolved into TAM2 [21] through the incorporation of social, organizational and work-related factors – including social norms, image, job relevance and outcome quality - that contribute to the explanation of how technology can be accepted/made use of within organizations. Further, it evolved into TAM3 with the addition of psychological and competency-related factors (including self efficacy, anxiety and previous experience) that influence perceived ease of use. TAM has been continuously refined to provide a richer definition and description of user behavior in an increasingly technologically complex world.

Researchers are developing new variations of and iterations to the various TAM models, and researchers are continually working to develop new variables or variables appropriate for their particular area of study.

D. Technology-Organization-Environment (TOE) Framework

The Tornatzky and Fleischer (1990) technology-organizational-environment (TOE) framework provides researchers a means of understanding the different factors that can influence technology adoption by organizations. The TOE framework uses three dimensions (technological, organizational, and external environment) to assess the impact of technology on organizations. Each of these three dimensions will influence both an organization's ability to adopt innovative technologies, as well as the overall success that the organization has in the implementation and use of new technology.

The TOE framework provides researchers with a high degree of flexibility to allow researchers to choose the specific sub-factors that are applicable to a particular context. For example, [22] used the TOE framework to study the adoption of A.I. by small and medium enterprises (SMEs) and were able to determine the following factors as significant: 1) Technological (competitive advantage and value), 2) Compatibility (conformity with existing processes and culture), 3) Organizational (readiness and capability of human resources), and 4) Environmental (market demand, external pressures, and support from the Government).

In addition to leveraging the flexibility of the TOE framework, the researchers also conducted a literature review to identify the factors affecting the use of A.I. applications in the quality of audit work.

E. Related research

According to our literature review, technology is influencing the success of auditing in three primary dimensions: (1) External Environment — Regulations are more significant than market mechanisms [17,26]; AI ethics considerations [6]; IT governance [27]; and Competitive

Pressures [24,28,29]. (2) Organization — there will need to be more employees with the requisite knowledge/skills to use AI [10,11,17]; Learning Culture is important [23]; Executive Support [24]; and Digital Strategy must match the size of the company [25, 26]; (3) Technology — investments made on AI and automation have produced reductions in both costs [10, 30] and real-time errors in audits [6, 13, 24]. At the same time, organizations must also be concerned about data security and technological complexity [15, 29].

This research creates a causal model by integrating the TOE and TAM frameworks to investigate the impact of technology, organization and environment (TOE) on the perceived value of AI (TAM) and how this impacts the users' behaviour and audit quality. The research is a quantitative study of survey data from auditors at medium-sized and small-sized firms in Thailand. The research applies the methodology of Badghish & Soomro [22], and uses the TOE framework to study the adoption of AI by SMEs. This methodology offers researchers great flexibility, as it allows them to change the variables to fit the context of the study, as shown in Fig.1.

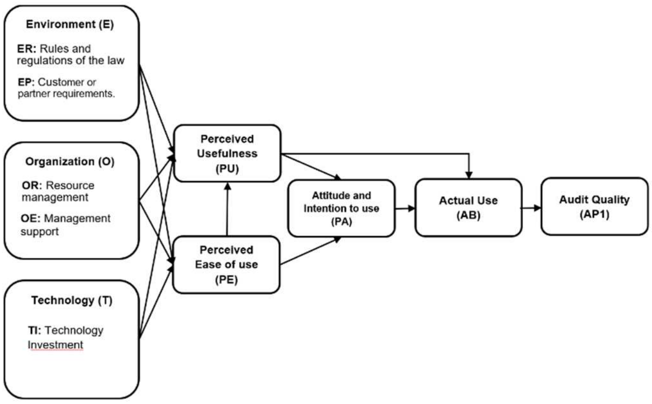

Fig. 1 Research Framework

III. RESEARCH METHODOLOGY

The research process is divided into three main parts: 1) conducting a literature review and developing a measurement tool; 2) testing and collecting data; and 3) analyzing the data collected, drawing conclusions based on the findings, and disseminating the results of the study.

A. Population and Sampling

The hypotheses are divided into 3 groups according to the relationship paths in the m In this research, the population (part of the overall study) and the sample (part of the population being studied) will consist of all accounting professionals working for small to medium sized accounting firms in Thailand (auditors, assistants, team leaders, and partners). Small to medium-sized firms are classified according to the criteria established by The Office of the Small and Medium Enterprise Promotion [31] and the Ministerial Regulation of Small to Medium-Sized Firms, B.E. 2562 (2019). The criteria for classification include (1) total revenue as reflected in the financial statements, and (2) number of employees. In the event that business size is indicated by both the number of employees and total revenue, the total revenue will be used as the first criteria to determine size, as shown in Table I.

TABLE I. THE TABLE SHOWS THE SIZE OF AUDITING FIRMS

Sizes of auditing firms	Annual income (million baht)	Employment (people)
Medium	Over 50 but not exceeding 300	Over 30 but not exceeding 100
Small	Over 1.8 but not exceeding 50	Over 5 but not exceeding 30
Sub-sized	Not exceeding 1.8	Not exceeding 5

According to Cohen J. (1992), a minimum of 91 participants were required for multiple regression analysis ($\propto = 0.05$, $\beta = .80$, five independent variable opinions); for the PLS-SEM method requiring the use of Hair et al., [32], a minimum of 120 actual participants provided a high standard of reliability for the opinions offered.

B. Research hypothesis

The hypotheses are divided into 3 groups according to Table II, which shows the relationship paths in the model as shown in Fig.1.

TABLE II. RESEARCH HYPOTHESIS

Group 1: Factors influencing auditors' acceptance of AI technology.	
H1	Regulatory and legal readiness (ER) has a positive influence on the perceived usefulness of AI (PU)
H2	Regulatory and legal readiness (ER) has a positive influence on the perceived ease of use of AI (PE)
H3	Customer or partner needs (EP) have a positive influence on the perceived usefulness of AI (PU)
H4	Customer or partner needs (EP) have a positive influence on the perceived ease of use of AI (PE)
H5	Resource availability (OR) has a positive influence on the perceived usefulness of AI (PU)
H6	Resource availability (OR) has a positive influence on the perceived ease of use of AI (PE)
H7	Management support (OE) has a positive influence on the perceived usefulness of AI (PU)
H8	Management support (OE) has a positive influence on the perceived ease of use of AI (PE)
H9	Technology readiness (TI) has a positive influence on the perceived usefulness of AI (PU)
H10	Technology readiness (TI) has a positive influence on the perceived ease of use of AI (PE)
Group 2: Relationship based on Technology Acceptance Theory and Actual Usage Behavior	
H11	Perceived ease of use (PE) has a positive influence on perceived usefulness of AI (PU)
H12	Perceived ease of use (PE) has a positive influence on AI acceptance and use (PA)
H13	Perceived usefulness (PU) has a positive influence on AI acceptance and use (PA)
H14	Perceived usefulness (PU) has a positive influence on actual implementation (AB)
H15	AI acceptance and use (PA) has a positive influence on actual implementation (AB)
Group 3: Analysis of the influence of AI use on audit quality	
H16	Actual implementation (AB) has a positive influence on audit quality (AP1)

C. Research Instrument

Using the information gathered from various authors' research articles, a survey was developed as the research instrument for this study to measure the influence of several variables on improving the auditing process. The survey was divided into seven parts: Demographics, Environmental Factors, Organizational Factors, Technology Factors, Perceptions Based on the Technology Acceptance Model (benefits, ease of use, and intention to use AI), Actual AI Usage Behaviour, and Quality of Audit Work. Additionally, the survey gathered respondents' opinions using a 5-point Likert Scale (from strongly agree to strongly disagree).

D. Development of Questionnaire

Using a literature review of the pertinent variables that compose this study, researchers formulated a set of questions based on the following categories: Environmental Factors (Legal Impact ER1, Government Policy ER2, AI Ethics ER3, Customer Needs EP1, Customer Size EP2, and Partner Pressure EP3) [17,22,26]; Organizational Factors (IT Knowledge OR1, AI Expertise OR2, Customer Industry Knowledge OR3, Executive Support OE1, Infrastructure & Budget OE2, and Innovation Policy OE3) [10,11,17]; Technological Factors (Ongoing Investments in AI TI1, System's Budget TI2, and Training TI3) [22]; Perceived Benefits (Improved Efficiency PU1, Reduced Errors PU2, and Fast Decision Making PU3) [24,28]; Perceived Ease of Use (Easy to use AI PE1, Quick to use PE2, and Uncomplicated to use PE3,) [6,14,23]; Attitudes toward AI (Confidence in Use PA1, Willingness to Support PA2, and Belief in Future of AI) [13,15,26]; Actual Use Behavior (Risk Screening AB1, Evidence Sampling AB2, Software Integration AB3, and Evaluation AB4,) [26]; Quality of Validity of Conclusions.

E. Data Analysis

The researchers coded and analyzed the data from the questionnaires prior to conducting a statistical analysis of how to provide demographic information and descriptions of variables in the following six areas: environment, organization, technology, acceptance/use of the technology, inspection quality. The researchers used two forms of statistical analysis: one to do an overall descriptive analysis of all six variable categories, and the second to utilize structural equation modeling with PLS-SEM via ADANCO software. To complete the statistical analysis, the first step was assessing the measurement model to determine if the measurement (survey) instrument had qualified based on reliability criteria: >0.70 (Cronbach's Alpha, Jöreskog's Rho, Dijkstra-Henseler's Rho), aggregate Validity >0.50 (AVE), and Tertiary Validity <0.85 (HTMT) [32]. Finally, the researchers assessed the structural models concerning their research hypotheses based on the coefficients associated with path analysis (β); the strength of their statistical significance (ρ-value) using bootstrap results; the strength of the relationship between the variables within the model's structure (R^2); and the overall strength +/or effects of using the technology to inspect something (f^2).

IV. Results

A. Sample Characteristics

A large proportion of the sample was made up of women (80.83%); the majority were aged between 45 and 54 years of age (40.83%) and are likely to be technologically literate. Approximately 70% of this population held a bachelor's degree. The most common job title among these participants was assistant auditor (55%); followed by auditor, (24.17%)*. The lowest number of participants were business partners (3.33%); and all were part of the target group. Almost all of this population had been working within the auditing profession for more than 10 years (40.83%); and suggest a good understanding of their jobs. The majority of the companies represented were 1.8M - 50M Baht in revenue with 5 - 30 employees (44.17%), as shown in Table III.

TABLE III. Sample Demographics

Personal Characteristics	Number (people)	Percentage
1. Gender		
Male	23	19.17
Female	97	80.83
2. Age		
Under 25 years	9	7.50
25–34 years	24	20.00
35–44 years	24	20.00
45–54 years	49	40.83
55 years and above	14	11.67
3. Highest Education Level		
Bachelor's Degree	90	75.00
Master's Degree	24	20.00
Doctorate	1	0.83
Other	5	4.17
4. Current Position		
Audit Manager	13	10.83
Audit Assistant	66	55.00
Auditor	25	20.84*
Partner in Audit Firm	12	10.00
Partner in Audit Firm (Auditor)	4	3.33*
5. Audit Experience		
Less than 3 years	37	30.83
3–5 years	14	11.67
6–10 years	20	16.67
More than 10 years	49	40.83
6. Size of Audit Firm		
Annual revenue exceeding 1.8 million baht but not exceeding 50 million baht and employing more than 5 but not more than 30 people	53	44.17
Annual revenue exceeding 50 million baht but not exceeding 300 million baht and employing more than 30 but not more than 100 people	28	23.33
Annual revenue not exceeding 1.8 million baht and employing no more than 5 people	39	32.50

B. Measurement Model Assessment

Data was analysed with ADANCO software using PLS SEM analysis, and the model displayed reliability and the ability to accurately demonstrate all established metrics (Cronbach's Alpha 0.818 – 0.957 Benchmark > 0.7, CR 0.818 – 1.000 Benchmark > 0.7, AVE 0.603 – 0.882 Benchmark > 0.5, HTMT 0.496 – 0.826 Benchmark < 0.85 ; Henseler et al., 2016). The independent variables identified had significant levels of predictive power (R^2) towards perceived usefulness (PU) (73.73%) and positive acceptability of AI (PA) (71.24%). Whereas, the variables ease of use (PE) (52.71%), applicable benefits (AB) (55.78%) and adequate quality for verification (AP) (44.13%) all exhibited moderate levels of predictive power. All hypothesis testing results and all variable relationships can be found in Figure 2.

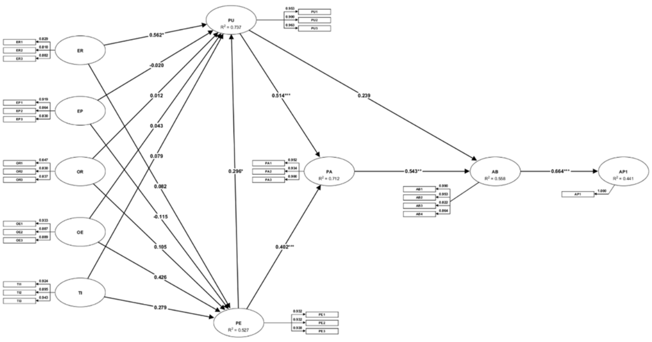

Fig. 2 Structure model and hypothesis testing

C. Hypothesis Testing Results

The analysis of path coefficients indicated that perceived benefit of AI usage was most influenced by regulatory and environmental factors, which had an overall positive correlation ($\beta = 0.562$; $p < 0.05$). In contrast, testing of the TAM theory found that perceived benefit (PU) was the primary factor for determining intention to use AI ($\beta = 0.514$) and was significantly greater than ease of use (PE). Furthermore, intention to use (PA) was found to have a significant positive effect on actual behaviour towards AI (AB) ($\beta = 0.543$), and accounted for 71.2% ($R^2 = 0.712$) of the variance [32]; thus indicating how strong the relationship was between intention and behaviour. Actual usage behaviour (AB) was also found to have a very large positive relationship with verification quality (AP1) ($f^2 = 0.790$), indicating how critical usage behaviour is for determining verification quality [33].

TABLE IV. HYPOTHESIS TESTING RESULTS

Factor	Relationship Path)Path)	Path Coefficient (β)	Cohen's (f^2)	T-Values	P-Values	Test results
Factors influencing auditors' acceptance of AI technology.						
Environmental factors						
Regulations and legal	H1: ER -> PU	0.562	0.323	2.037	.042*	**Accept**
requirements	H2: ER -> PE	0.082	0.004	0.277	.782	Reject
Customer or partner needs	H3: EP -> PU	-0.020	0.000	-0.099	.922	Reject
	H4: EP -> PE	-0.115	0.008	-0.477	.633	Reject
Organizational factors						
Resource management	H5: OR -> PU	0.012	0.000	0.053	.958	Reject
	H6: OR -> PE	0.105	0.012	0.411	.681	Reject
Management support	H7: OE -> PU	0.043	0.001	0.087	.931	Reject
	H8: OE -> PE	0.426	0.065	0.981	.327	Reject
Technological factors						
Technology investment	H9: TI -> PU	0.079	0.007	0.350	.763	Reject
	H10: TI -> PE	0.279	0.048	1.201	.230	Reject
The relationship between technology acceptance model and actual usage behavior.						
Perceived ease of use	H11: PE -> PU	0.296	0.158	2.475	.013*	**Accept**
	H12: PE -> PA	0.402	0.291	3.898	.000***	**Accept**
Perceived usefulness	H13: PU -> PA	0.514	0.478	4.714	.000***	**Accept**
	H14: PU -> AB	0.239	0.048	1.453	.146	Reject
Actual AI usage behavior	H15: PA -> AB	0.543	0.248	2.924	.004**	**Accept**
Analyzing the influence of AI on the quality of audit work.						
Help use technology to	H16: AB -> AP1	0.664	0.790	8.384	.000***	**Accept**
Increase the accuracy of audit results						

** is statistically significant at the .05 level (t-value > 1.96), ** is statistically significant at the .01 level (t-value > 2.58), and *** is statistically significant at the .001 level (t-value > 3.29).*

V. DISCUSSION AND RECOMMENDATIONS

A. Discussion

The use of AI in auditing has been impacted by government regulations, policies, and ethics as evidenced by the research conducted by both Lombardi et al. (2025) [17]. Auditors tend to place more weight on the perceived benefits (or advantages) that a solution offers an auditor than how easy it would be to use (e.g., user-friendliness), as demonstrated in research conducted by Mansour et al. (2025) [23]. Auditors believe that using AI enables them to perform higher quality audit procedures, or, in other words, will generate superior audit outcomes, as demonstrated in research conducted by both Black & Gerard (2025) [13] and Foehr et al. (2025) [15].

B. Research Limitations

There are a number of limitations to this study: 1) The study's population consists only of auditors from small and medium-sized enterprises (SMEs) from Thailand, therefore it will be difficult to compare to large accounting firms (big four) or international firms; 2) The sample size of 120 respondents is sufficient for statistical analysis with PLS-SEM; however, the population findings should be viewed with caution before making generalised conclusions; 3) This is a cross-sectional study that was conducted at one point in time, meaning that opinions can and will change as AI continues to develop; 4) TOA and TAM may not capture all relevant determinants of AI acceptance, and therefore future researchers will benefit from considering additional variables outside of the TOE and TAM frameworks.

C. Future Research Recommendations

According to this study, audit firms need to include AI as part of their day-to-day operations. In order for digital transformation within audit departments to be successful, the management of firms will need to actively support and train their personnel utilizing AI. Professional organizations should create various standardizations and resources to help firms close the gap between technology and audit. Future research should examine the ethical implications of using advanced technology, such as generative AI, from those who have refused to do so using qualitative methodologies, while also assessing client confidence and opinion towards AI-generated financial statements to gain a complete understanding of the situation.

VI. CONCLUSION

According to empirical research conducted on Thai SME auditors, the biggest environmental factors affecting an auditor's use of AI are laws & regulations followed by government support policy and ethics surrounding AI Use. Technology acceptance theory was confirmed based on this research as there were no significant organizational or technological factors (skills or budget) affecting auditor's use of AI. The results also indicate that perceived benefits from using AI are more important to auditors than the ease at which they can use AI, as demonstrated through improved quality of auditing. Additionally, this research provides theoretical benefit through the identification of the influence of laws, ethics and government policy on the decision of auditors to use AI, and practical benefit through demonstrating the need for accounting firms to adequately prepare themselves with their management leading the AI Strategy, focusing on upskilling their staff, while the Federation of Accounting Professions should create standard guidelines and provide common tools for bridging the technological gap.

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